

DIFFERENT COLOURS OF THE TRANSITION METALS

COPPER (II), Cu^{2+} - BLUE

IRON (II), Fe^{2+} - PALE GREEN

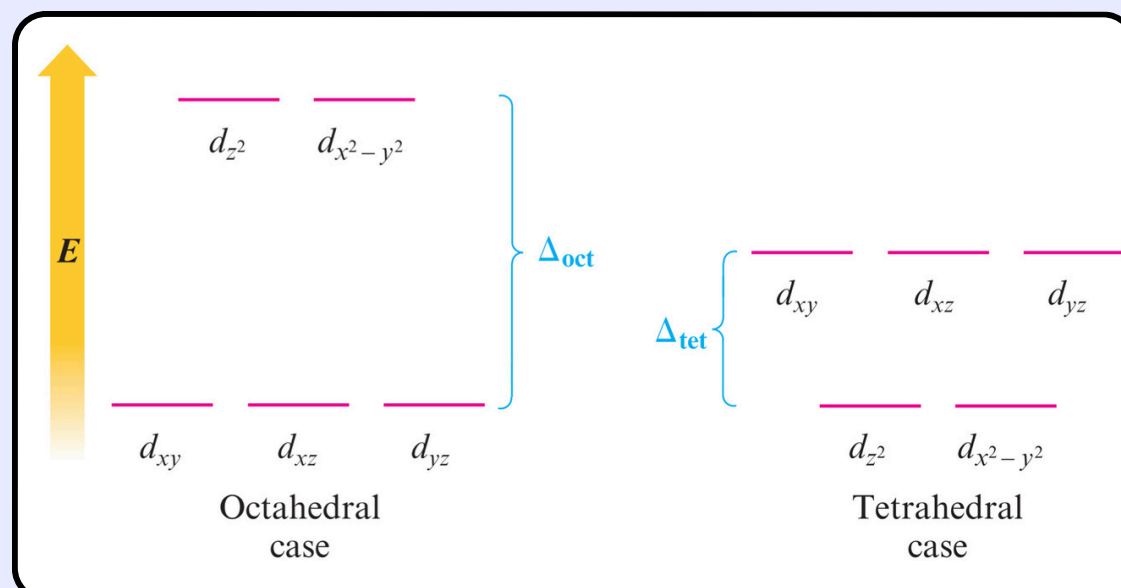
IRON (III), Fe^{3+} - YELLOW / BROWN

COBALT (II), Co^{2+} - PINK

D-ORBITALS

Transition metals have d orbitals. When the metal ion is isolated, all five d orbitals have the same energy.

But in real chemistry, metal ions are surrounded by ligands (H_2O , NH_3 , Cl^- , CN^-). These ligands repel the d electrons unevenly, causing energy splitting.

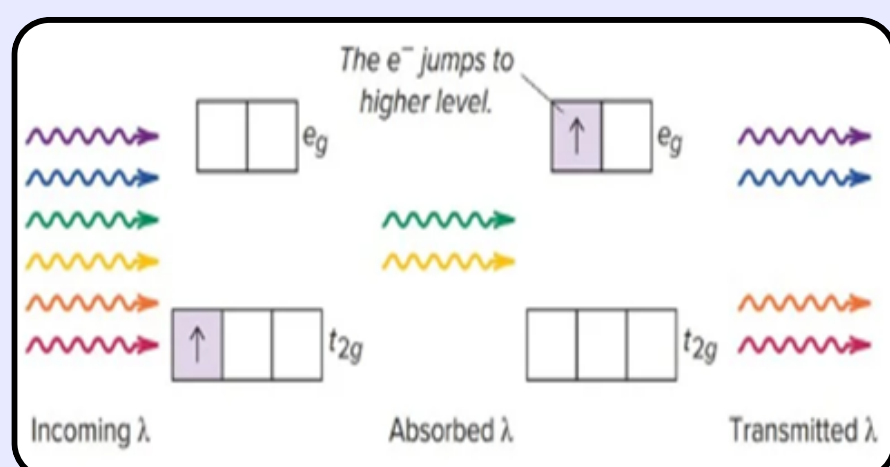


CRYSTAL FIELD (OR LIGAND FIELD) SPLITTING

PRESENTED BY GROUP 4A

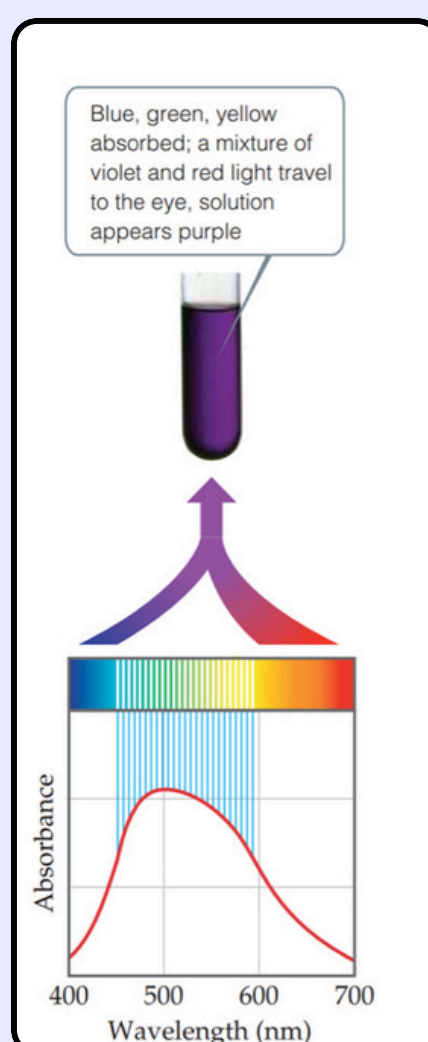
TRANSITION METALS' COLOURS

ENERGY GAP



When **VISIBLE** light is shone, electrons in the lower-energy sets, jump up to the higher-energy set. The energy needed to do this is the **ENERGY GAP**.

The colour transmitted by the metal has its energy correspond to the energy gap.



SUMMARY

“Transition metals are colourful because their partially filled d-orbitals split into energy levels that interact perfectly with visible light, turning tiny electronic jumps into stunning colours.”

